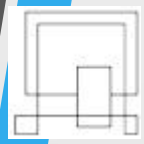
VIA University College



Programming 3

Autumn 2026

Learning Objectives

- By the end of this session, you should be able to:
  - ✓ Explain Spring Data JPA
  - ✓ Explain Spring Data Repositories Interfaces
  - ✓ add Spring Data JPA to Spring Boot
  - ✓ Explore Domain/Entity annotations

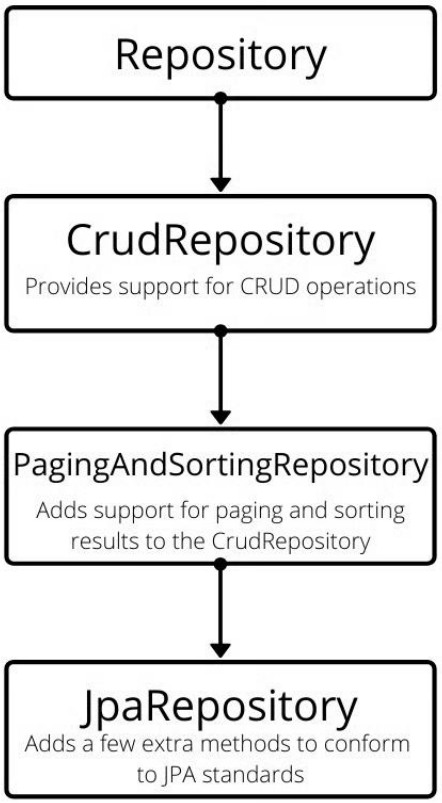
What is Spring Data JPA?



- Spring Data Java Persistence API (JPA) is used to reduce the amount of boilerplate code required to implement the data access object (DAO) layer.
  - It includes both relational databases systems and NoSQL data stores.
- It supports advanced integration with Spring MVC controllers.
- It provides support for creating JPA repositories by extending the Spring Data repository interfaces.

Spring Data Repositories Interfaces

- The power of Spring Data JPA lies in the repository abstraction (provided by the Spring Data Commons project)
- We can use Spring Data JPA without paying attention to the actual implementation of the repository abstraction



https://codingnomads.com/spring-data-jpa-repository

Adding Spring Data JPA to Spring Boot

- We need to configure the required dependencies in the pom.xml file.
- Add dependencies:

```
<!—Database (PostgreSQL) -->
<dependency>
    <groupId>org.postgresql</groupId>
    <artifactId>postgresql</artifactId>
    <scope>runtime</scope>
</dependency>

<-- Spring Data JPA -->
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```

Dependencies - MySQL

- Add dependencies:

```
<!—Database (MySQL) -->
<dependency>
    <groupId>com.mysql</groupId>
    <artifactId>mysql-connector-j</artifactId>
    <scope>runtime</scope>
</dependency>

<-- Spring Data JPA -->
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```

Query Methods

- Query methods are methods that find information from the database and are declared on the repository interface.
- Example:
  - To create a database query that finds the Vinyl object that has a specific name

```
public interface VinylRepository extends JpaRepository<Vinyl, Long> {
    Vinyl findByName(String name);
}
```

Creating the properties File

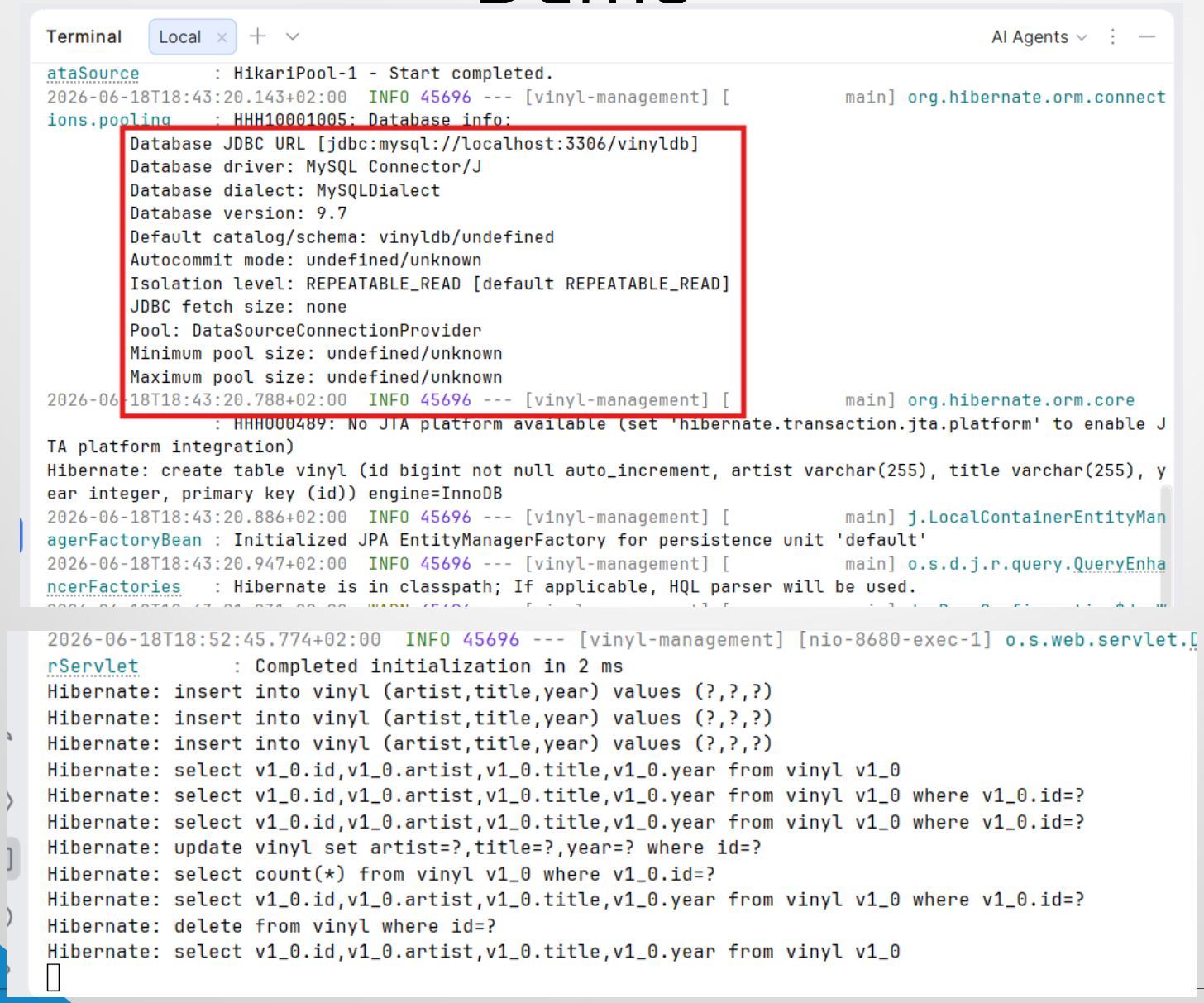
- The application.properties file contains the configuration used to configure our application consisting of:
  - Database connection: the name of the JDBC driver class, the JDBC url, the username of the database as well as the password
  - Hibernate-: database dialect, etc.
- Example:

```
# Postgres database configuration
spring.datasource.url=jdbc:postgresql://127.0.0.1:5432/dbname
spring.datasource.username=your_database-user
spring.datasource.password=your_secret_password
spring.datasource.driver-class-name=org.postgresql.Driver

# JPA, Hibernate configurations
spring.jpa.database-platform=org.hibernate.dialect.PostgreSQLDialect
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
```



Demo


Terminal
Local
AI Agents
ataSource : HikariPool-1 - Start completed.
2026-06-18T18:43:20.143+02:00 INFO 45696 --- [vinyl-management] [main] org.hibernate.orm.connect
ions.pooling : HHH10001005: Database info:
Database JDBC URL [jdbc:mysql://localhost:3306/vinyldb]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 9.7
Default catalog/schema: vinyldb/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-06-18T18:43:20.788+02:00 INFO 45696 --- [vinyl-management] [main] org.hibernate.orm.core
: HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable J
TA platform integration)
Hibernate: create table vinyl (id bigint not null auto_increment, artist varchar(255), title varchar(255), y
ear integer, primary key (id)) engine=InnoDB
2026-06-18T18:43:20.886+02:00 INFO 45696 --- [vinyl-management] [main] j.LocalContainerEntityMan
agerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-06-18T18:43:20.947+02:00 INFO 45696 --- [vinyl-management] [main] o.s.d.j.r.query.QueryEnha
ncerFactories : Hibernate is in classpath; If applicable, HQL parser will be used.
2026-06-18T18:52:45.774+02:00 INFO 45696 --- [vinyl-management] [nio-8680-exec-1] o.s.web.servlet.D
rServlet : Completed initialization in 2 ms
Hibernate: insert into vinyl (artist,title,year) values (?,?,?)
Hibernate: insert into vinyl (artist,title,year) values (?,?,?)
Hibernate: insert into vinyl (artist,title,year) values (?,?,?)
Hibernate: select v1_0.id,v1_0.artist,v1_0.title,v1_0.year from vinyl v1_0
Hibernate: select v1_0.id,v1_0.artist,v1_0.title,v1_0.year from vinyl v1_0 where v1_0.id=?
Hibernate: select v1_0.id,v1_0.artist,v1_0.title,v1_0.year from vinyl v1_0 where v1_0.id=?
Hibernate: update vinyl set artist=?,title=?,year=? where id=?
Hibernate: select count(*) from vinyl v1_0 where v1_0.id=?
Hibernate: select v1_0.id,v1_0.artist,v1_0.title,v1_0.year from vinyl v1_0 where v1_0.id=?
Hibernate: delete from vinyl where id=?
Hibernate: select v1_0.id,v1_0.artist,v1_0.title,v1_0.year from vinyl v1_0

Exercises

